Institute of Geographical Information Systems

CS-250 - Data Structures and Algorithms LAB

Semester: Spring 2026	**Class: SCEE-IGIS - 2024**
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LAB 9: Exploring Binary Trees in Data Structures & Analysis

Task 1: Build a Static Binary Tree

- Manually create a binary tree of at least 7 nodes using a predefined structure in your code (do not take input from the user).
- Each node should store a character (e.g., A, B, C, etc.).
- Display the structure using an inorder traversal.
- This task helps you practice tree declaration and traversal using a static tree.

Task 2: Implement Tree Traversals

- Inorder Traversal (Left, Root, Right)
- Preorder Traversal (Root, Left, Right)
- Postorder Traversal (Left, Right, Root)

Task 3: Search for an Element

- Write a recursive function to search for an element in the binary tree.

Task 4: Calculate Height of Tree

- Implement a function that returns the height of the binary tree.

Task 5 (Advanced): Count Leaf Nodes and Total Nodes

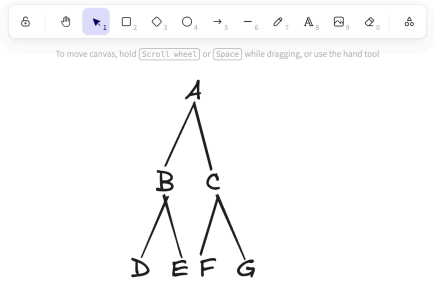
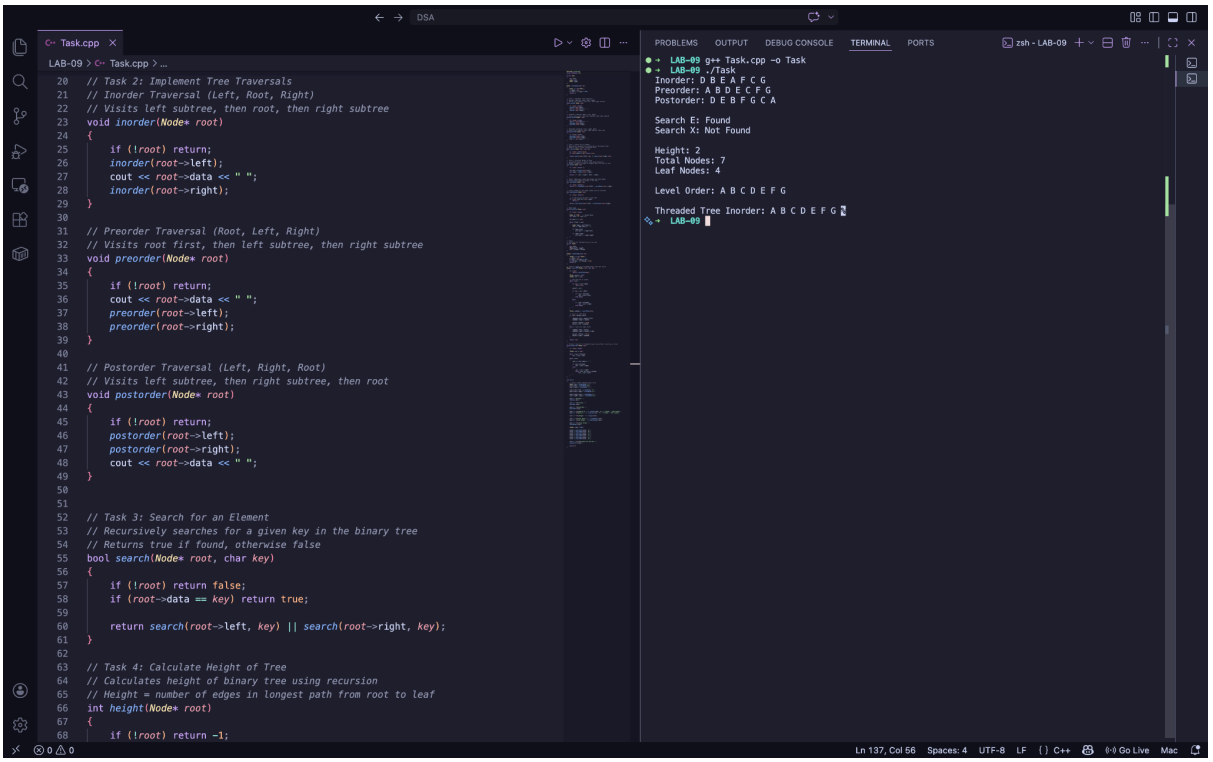
- Count the number of leaf nodes and total nodes in the binary tree.

Task 6

- Implement a threaded binary tree where null pointers are replaced with pointers to inorder predecessor and successor. Create functions to:
- Insert nodes in the threaded binary tree.
- Traverse the tree using inorder traversal without recursion or stack.
- Display the threading behavior with sample data.

Bonus Task (Optional):

- Visualize the tree using level-order traversal.
- Implement iterative versions of the traversal functions using stacks.

Submission Instructions:

- Submit your `.cpp` file via LMS
- Include comments for each function.
- Submit screenshots of output for at least one custom tree input.